



Driving innovation in Building Digital Twins!



International Conference on Digital
Frontiers in Buildings and Infrastructure
(DFBI2025)



Future Standards on DTw in the Built Environment

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A BUREAU VERITAS COMPANY





What is the BDTA??
EU Non-profit created in
Antwerpen



VISION

To allow the coordination of a multicriteria efficient decision-taking across all AECOO stakeholders by digitally duplicating EU building stock.

MISSION

Develop an Open & Trustworthy Technoeconomic Ecosystem to advance towards full implementation of Building Digital Twin Environments across all the stakeholders and lifecycles of AECOO* Smart and Connected Real State and Infrastructure Assets, including occupants, through professional development and community.

**Architecture, Engineering, Construction, Owner & Operator*

GUIDING PRINCIPLES

GP 1. OPEN

Develop an Open Technology Framework ready to evolve as a community along the lifespan of AECOO assets and cities

GP 2. ETHICAL

Provide a Citizen Centered Vision to guarantee Privacy and Ethical usage of the generated data

GP 3. RELIABLE

Synchronize updated information of smart and connected buildings towards an un-siloed complete information

GP 4. CIRCULARITY

Allow seamless environmental assessment across lifecycles for a circular management of building environment

GP 5. CLIMATE GOALS

Optimize Energy Performance in buildings by reducing gap projections between lifecycles

GP 6. ECONOMIC VALUE

Generate trustworthy baselines to generate economical guarantees to current and new business in AECOO sector

BDTA STRATEGIC AXIS

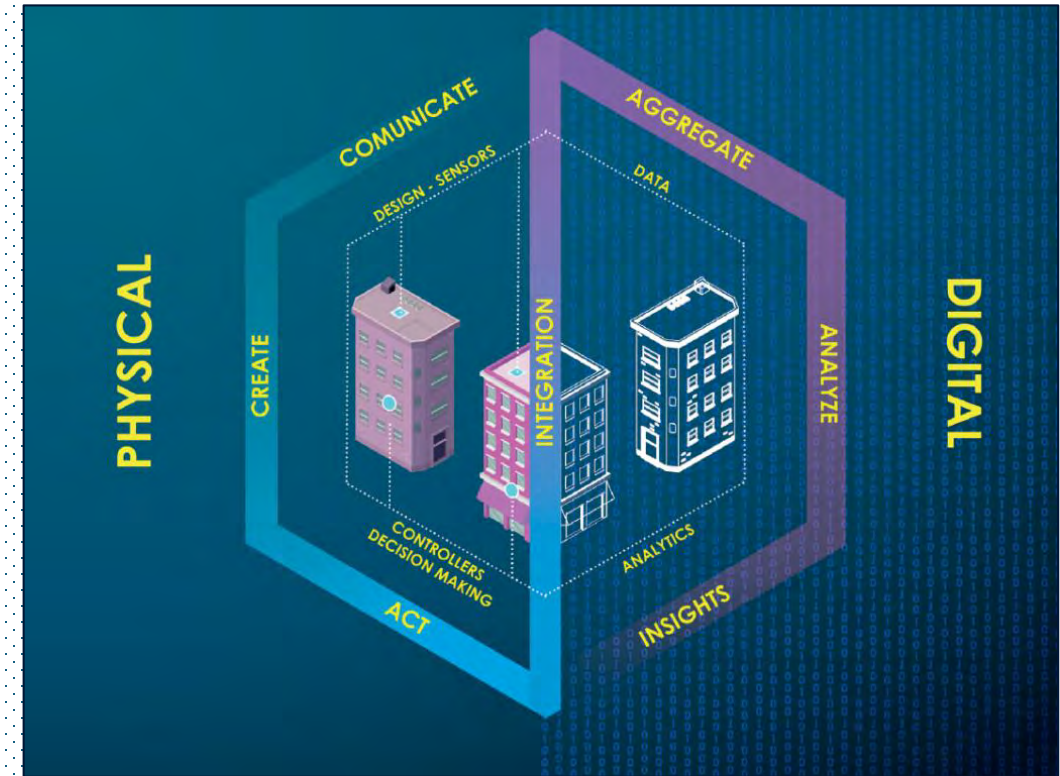
Community agreed developments in Processes, Skills and Tools!

METHODOLOGY: Definitions and New Roles

PUBLICATIONS: White Papers, Technology Insights, Standards

OPEN TECHNOLOGY: Data governance tools and best practices

BDTw COMMUNITY: Events and Working groups



BDTA WORKING GROUPS

WG1. DEFINITIONS AND ROLES → Standardization roadmap, trainings and certifications

WG2. DATA INTEROPERABILITY → Ontology Network Catalog development for Data Reference Architecture in BDTw

WG3. SIMULATION ACROSS LIFECYCLE → Simulation Model Tracking System (STMS) and SIMBots

WG4. ETHICAL AND PRIVACY METRICS → Trustworthiness assessment

WG5. CIVIL INFRASTRUCTURES → Dedicated Use Cases & Vertical Aggregation

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Ambition: To generate a community based common set of definitions, roles and their related procedures, to reach interoperable Building Digital Twins Environments (*Interrogative and Predictive Purpose*).

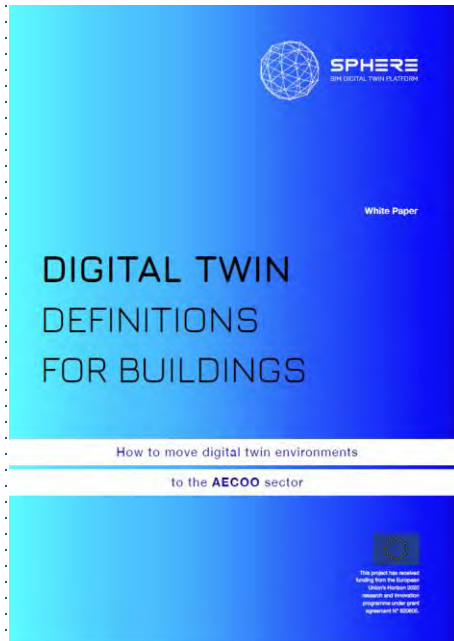
A set of **Building Digital Twin Definitions** generated under the SPHERE research were collected as an open White Paper in Q4 2019.



SPHERE project has received funding from the European Union's H2020 programme under Grant Agreement No. 820805.

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

2019 BDTA whitpaper



The DT development in the manufacturing industry is underway in parallel and with learnings from a highly related digital manufacturing technology called cyber-physical systems (CPS), a term first coined in 2006 by Helen Giller from the US National Science Foundation (Lee and Seshia 2017).

The concept of a cyber-physical system is a manufacturing system where both physical components and their virtual counterparts or extensions are integrated using sensors, controllers, and computer algorithms, such that the physical space can be monitored, coordinated and controlled by the virtual side in an automated manner a seamless manner (Wang et al. 2015).

Cyber-physical systems and Digital Twins are similar in that they have physical-digital connection, near real-time interaction, integration of organisational management across an asset's life cycle, and providing the means to enhance collaboration, yet they differ in their level of representation of the asset. Cyber-Physical systems focus on computing communication, and control (CC) primarily for automation and the cyber system consists of algorithms not of a digital representation.

CPS require a data stream of controller or sensor inputs from the physical asset, algorithms that provide computations, and their outputs communicated to control the physical system, with potential feedback loops. The computational algorithms can be of any kind, whereas the algorithms do not need to feature any digital representation of the asset itself.

In contrast, Digital Twin's focus on a virtual representation of a physical asset in relevant forms that can provide for simulation on top of real-life data, and operational management decision support (Tao et al. 2019). As such a CPS can be, but does not have to be, part of a Digital Twin and when it does, it is typically referred to as being instantiated through CPS twinning (Damjanovic & Behrendt 2019). DT's and CPS are thus emerging as two separate yet linkable digital transformation R&D efforts.

5. NEW DEFINITIONS OF BUILDING DIGITAL TWINS FOR AECOO FROM THE SPHERE PROJECT

Whilst the Design and Construction phase have usually been the ultimate purpose of all AECOO's stakeholders, a change is currently occurring connecting works carried out on buildings under "design, manufacturing, and construction" with their future operational performance, extended AECOO to the AECOO sector (Architecture, Engineering, Construction, Ownership AND Operation). This connection across life phases is the ultimate mission of current Product Life Cycle Management systems and tools under development.

Under the SPHERE project (www.sphere-project.eu, Grant Agreement No. 82098) a set of definitions have been developed by the consortium to help drive the technical Digital Twin development for the AECOO sector, forming the core of this White Paper. The definitions help to provide a common understanding both on what provides for a Building Digital Twin, and the deeper knowledge on how to implement it, to create a common definition landscape across stakeholders in the AECOO sector.

The definitions development fit with the technical objective of the SPHERE H2020 EU project which is to develop an IoT platform able to manage and update both static and dynamic building information across the design, construction/renovation and operations phases from multiple sources, and utilise this information both in operational energy control, and for feed-backs from operations to design.

Beyond SPHERE, the definitions proposed below are aligned with the same concept raised by Digital Built Britain mentioned in the previous section, yet expand their scope to fill the existing gap between BIM current definitions and procedures with the completion of a real Digital Twin Environment for Buildings as already known in other Manufacturing Sectors.

All the new definitions and related procedures developed in SPHERE, as shared below forming an initial definition landscape, serve to provide a set of implementable Building Digital Twin definitions that will be tested in 4 real AECOO pilots across the EU (including New buildings and major retrofitting projects in existing ones) in the following 3 next years.

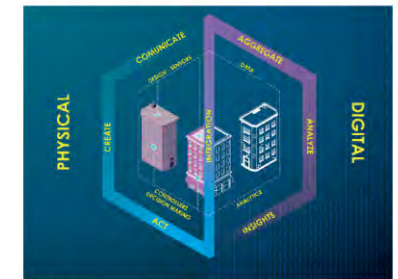


Figure 6: The Digital Twin concept. Data are captured and streamed to a digital platform, which, in turn, performs real-time analysis to optimise the design and the performance (adapted by M. Elmaghrabi based on Deloitte University Press).

Digital Twin definitions for new buildings

As such, the following definitions as per Exhibit 8 have been developed for Digital Twins of new buildings to initiate a common definition landscape.

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

2019 BDTA whitepaper



EXHIBIT 8. PROPOSED NEW DEFINITIONS LINKING BIM TO DIGITAL TWINS FROM SPHERE

A Building Digital Twin describing the AECOO asset during its design and construction. It contains the informational sets necessary to describe and produce a physical version that duplicates or twins the virtual version. These informational sets include, but are not limited to:

- Employer Requirements included inside the contracts, previous legal and geotechnical information of the land,
- An Information Model in the form of an Object Type DB which will include 3D Geometrical Information as well as Materials and Elements Inventory,
- Bill of Processes as a comprehensive log of past processes and actions based on Minutes and Confirmed communications,
- Bill of Services as the current evolution on the functional characteristics of the asset, starting back from initial Employer Requirements,
- Bill of Disposal (e.g. Material Passports, Selected Deconstruction Techniques, Circular Construction...)

BDTP evolution are crucial for both via iterative improvements through Model Simulations as well as for the usage of their associated values as baseline of DTIs after the commissioning of a real building asset.

Building Digital Twin Prototype (BDTP)

Building Digital Twin Instance (BDTI)

Building Digital Twin Aggregate (BDTA)

A Building Digital Twin that is linked to throughout the life of a specific corresponding physical product, including pairing when the asset starts its operation. In the AECOO sector, this milestone corresponds to the legally binding As-Built Documentation. This type of Digital Twin may contain, but again is not limited to, the following information sets:

- A fully Information Model which will include the 3D model with General Dimensioning and Tolerances (GD&T) that describes the geometry of the physical instance and its components,
- Bill of Materials that lists current and previous elements and components (mostly relative to architecture, structural and building services),
- Bill of Process that lists the operations that were performed in creating the physical instance (based on the logs and project variations generated after the Rehearsing act), along with the results of any measurements and tests on the instance (e.g. Cloud Points Surveys of inner services, Structural load tests, GeoRadars...),
- A Service Record that describes past services performed and components replaced (Major/Minor Refitting and regular Upkeep),
- Operational States captured from actual sensor data, current, past actual, and future predicted (BMS, SCADA/IoT sensors, Facility Management Servers, Simulations...)

In AECOO, similar to manufacturing, this type of Building Digital Twin is the aggregation of multiple DTIs, instead of having an independent data structure. DTAs are a computing construct which provide direct access to all DTIs, hence allowing ad-hoc or proactively queries for benchmarking and comparisons.

This type of DT is following the same principle than in manufacturing use case, by linking similar DTIs. However, in general AECOO products presents a smaller number of assets and much more heterogeneity than manufacturing products. In addition, the legal persons involved in their production are many more from in manufacturing supply chains (e.g. OEMs and Tier1), hence providing a much more distributed data ownership. These two inherent characteristics of AECOO use cases may hinder both the representativeness of the aggregated data as well as the potential of queries scopes.

From the point of view of many related stakeholders of the AECOO sector (owners, tenants, Public Administrations, associations...), a real state (containing) has to be considered jointly as a container for the products held inside itself (contents) like appliances, furniture, and personal goods. These contents may, as well as important components of the building services and auxiliary systems (e.g. Buildings Automations), could have their own Digital Twin Instances. The connection among these multi-scale Digital Twin Instances will highly expand the boundaries of access to relevant information. On the other side, many actors are nowadays producing DTIs from Critical Civil Infrastructures and even whole districts are proposed, thus reaching infrastructure-scale and city-scale. Building scale is then the hinge in between cities and its contents, and the synchronization of Building Digital Twins across the Digital Twin scales will allow the vertical aggregation from appliances to districts. In there, vBDTA will put the focus on the AECOO stakeholders needs of relevant information and optimized queries.

Beyond this, by the creation of any DTE, two main purposes are sought: Predictive (intimately linked with simulation tools or DTIs and DTIs) and Interrogative (applying to DTIs as well as to DTIs in-depth analysis). As it happens to be in any DTE, these two basic drivers may include completely different requests depending on which is the role of the stakeholder interacting and the typology of the Digital Twin.

Manufacturing Building Digital Twin Aggregate (MBDTA)

Building Digital Twin Aggregate (BDTA)

Building Digital Twin Environment (BDTE)

EXHIBIT 9. PROPOSED NEW DEFINITIONS LINKING BIM TO DIGITAL TWINS FROM SPHERE

Using existing containers of information related to AECOO projects under the ISO 19650-1:2018 will bring most of the desired features to create a BDTP during design and construction phases. The Project Information Model content mainly on BIM Model developed inside the CDE will be the main source of the DTP; although not the only one, since BDTPs will include at least, although not limited to, a BEP legal document agreed by the main stakeholders involved in this phase, developing legal framework, standards, technical procedures, personnel, data property and the scope and all internal and external requirements (BIM-BAP), as well as the CDEs of the key organizations, previous files related to the setting, including terrain laser information (e.g. Cadastre, Historical...), as well as its geotechnical information (topological surveys, geotechnical campaigns...), it also includes other not structural data, conforming an Information Model. Also a PIM moved to a BDTP should include links between other databases related to the model which are more managed and updated dynamically. In relation to this managing the data of the PIM outside of the BIM Models itself is a challenge that BDTP should address.

BDTPs must integrate all the relevant information stored by the BEP and will hence sequentially use past Information Models to structure this type of Digital Twin. By considering again AECOO projects develop under the ISO 19650-1:2018, a new asset entering in its operation and maintenance phase may include store the evolution of the AM starting on from the previous As-Built Model of Information. This initial AM and its following recordings will hence provide the necessary information of the current instance including the General Dimensioning and Tolerances (GD&T), Bill of Materials and the Bill of Representative Processes (including activities in the BEP and BCP communications), along with the results of any measurements and tests on the instance (e.g. Cloud Points Surveys of inner services, Structural load tests, GeoRadars...), a Service Record that describes past services performed and components replaced (Major/Minor Refitting and regular Upkeep), and Operational States captured from actual sensor data, current, past actual, and future predicted (BMS, SCADA/IoT sensors, Facility Management Servers, Simulations...). All of this information must be linked dynamically to the BIM Models that compose the AM.

The computational construct of how to allow enough synchronization of an heterogeneous set of Digital Twins is the most underdeveloped field of the Building Digital Twins. However, there are some developments in current BIM SoA indirectly aiming to this concept, like the use of standardized formats in order to homogenize the whole aggregation of information. Moreover, according to ISO-19650, AIR could be the way to develop specifications that allow to manage future BDTP, because it implies a set of standardized definitions of the model objects used in the AMs. SPHERE project is currently proposing horizontal BDTP by the usage of a Platform as a Service Architecture which will include at least the following services:

- Linked Data Semantics Ontology,
- Integrated IoT Platform
- IFC Reader
- Block Chain
- Digital Twin Object Libraries

Assumptions based on the experience in manufacturing allow to forecast this will be the most rewarding by a complete multidisciplinary PIM assessment of horizontal and vertical Building-centered aggregations.

The BDTP will be able to expand any current CDE functionalities to be able to manage all the PIM and AM information dynamically, plus the connection with the on-site monitoring devices and actuators, especially for the asset operational stage.

BIM based Building Digital Twin Prototype

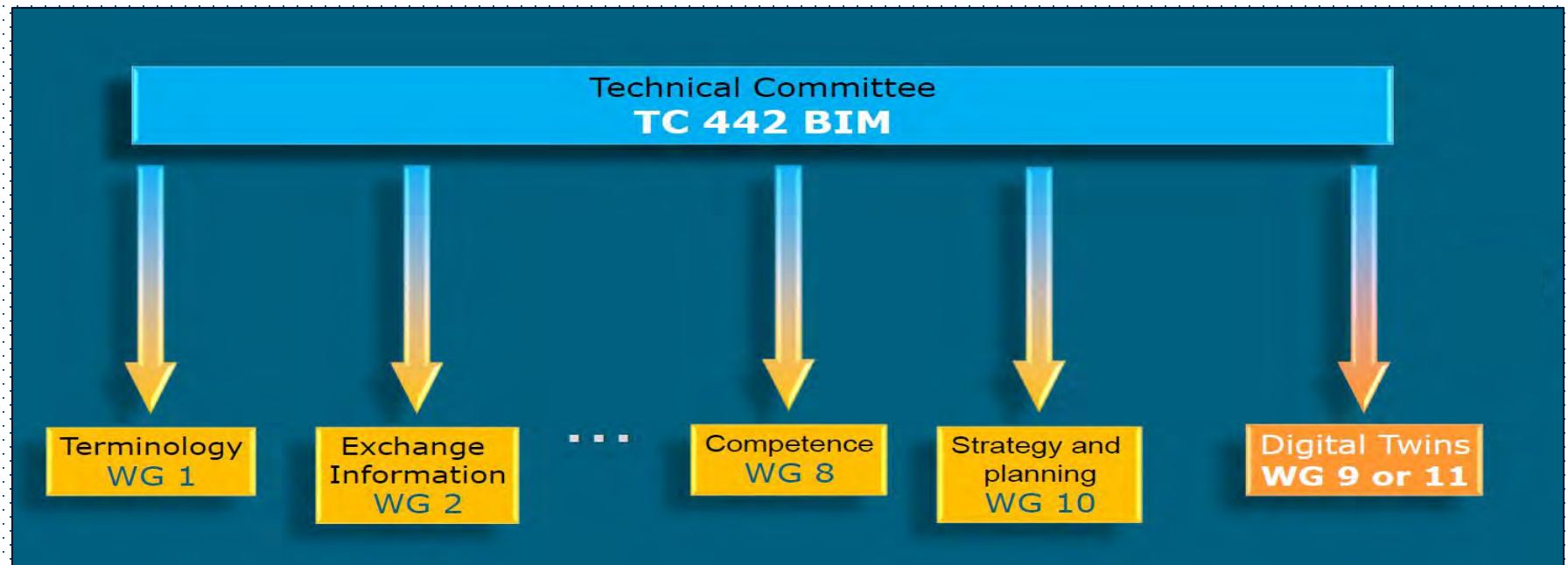
BIM based Building Digital Twin Instance

BIM based Building Digital Twin Environment

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

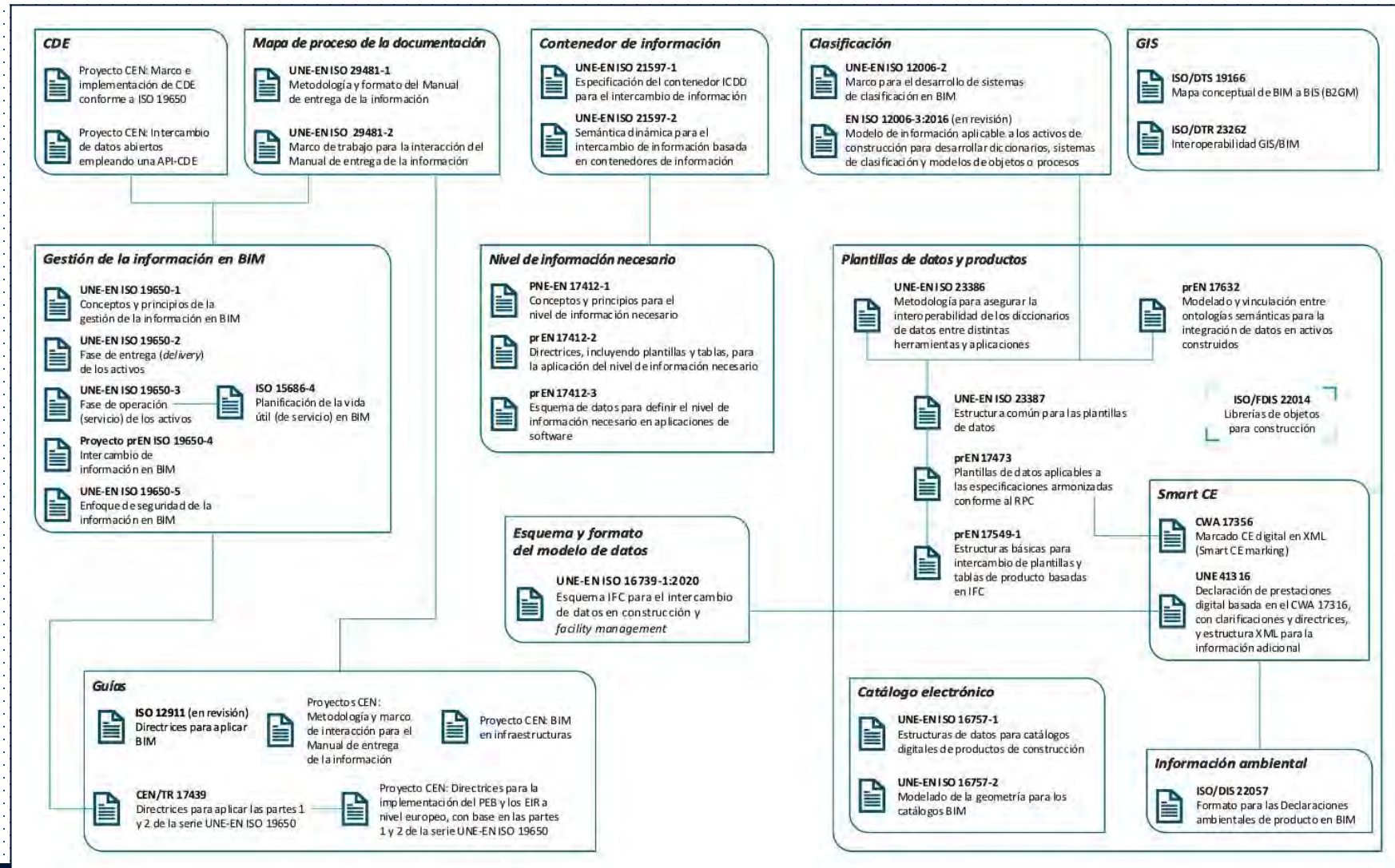
First Steps 2022 → Creation of the Working Group 9 (Building Digital Twins) under 442 Technical Committee

Secretariat:



WORKING GROUP #1: CEN Technical Committee 442 (BIM)

BIM norms map 2020
(source UNE)



WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442 (BIM) Structure

Working group	Title	Scope
CEN/TC 442/WG 1	Terminology	Provide advice to the TC 442 WG's on matters and issues related to BIM Terminology and harmonization of language across the standards developed by TC 442 Work in close cooperation with ISO/TC 59/SC 13/TF 01 Terminology
CEN/TC 442/WG 2	Exchange information	Develop and propose to TC442 when relevant, standards and guidelines to specify data structure for entities, geometry, and related properties as well as classification for exchanging data models. The data model ensures exchange of object-based information
CEN/TC 442/WG 3	Information Delivery Specification	Develop and propose to TC442 when relevant, standards and guidelines to specify how to describe the required information supporting a given process. Develop and propose to TC442 when relevant, standards and guidelines for Information Management using BIM

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442 (BIM) Structure

Working group	Title	Scope
CEN/TC 442/WG 4	Support Dictionaries	To develop and propose to TC442 when relevant, standards and guidelines to enable all stakeholders to operate using common machine-readable language by: - Providing and improving: - o methodologies to describe governance of dictionaries and their content; - o methodologies to describe construction objects based on data template and property dictionaries; - o methodologies to specify how to apply linked data and semantic web technologies. - Facilitating the WG4 standards implementation by providing guidelines for stakeholders; Investigating and facilitating the use of classification(s) systems by stakeholders

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442 (BIM) Structure

Working group	Title	Scope
CEN/TC 442/WG 5	Chair's Advisory Group	The Chairperson's Advisory Group, "CAG", is a permanent group created with the aim to coordinate and advise on generic issues across the CEN/TC 442 working groups. It provides a repository for information which is essential for the working groups during the drafting stages of their standards. The Chairperson's Advisory Group "CAG" does not develop standardization deliverables or makes formal decisions. The Advisory Group reports to CEN/TC 442. The Secretariat of the Advisory Group shall be the Secretary of CEN/TC 442. The Convenor shall be the Chairperson of CEN/TC 442.

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442 (BIM) Structure

Working group	Title	Scope
CEN/TC 442/WG 6	Infrastructure	Identify key stakeholders, investigate existing activities and formulate the need for standardization related to the implementation of BIM for infrastructure in Europe, not covered by existing standards and ongoing standards development. Make recommendation on whether standards should be developed and if so, how this should be done.
CEN/TC 442/WG 7	Horizontal role	Investigate and propose collaborative framework between CEN/TC 442 and other CEN and CENELEC TCs and standards committees with an interest in BIM methodologies and processes, To produce normative or informative documents to support the implementation of the horizontal role.
CEN/TC 442/WG 8	Competence	<i>Develop and propose to TC442 when relevant, standards and guidelines defining requirements of competence, knowledge and skills related to the Information Management enabled by the Information Modelling.</i>

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442 (BIM) Structure

Working group	Title	Scope
CEN/TC 442/WG 9	Digital twins in built environment	<i>Standardization in the field of structured information for the digital twins applied to the built environment, considering methodologies and formats to define, describe, exchange, monitor, record and securely handle digital twin's data and its related processes in relation with AECOO</i> <i>(Architecture, Engineering, Construction, Operation & Ownership).</i> <i>Digital twins applied to other industries, such as aerospace, cars or machines, are out of the scope of this proposal.</i>
CEN/TC 442/WG 10	Strategy and planning	

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442 (BIM) Structure

Working group	Title	Scope
CEN/TC 442/WG 11	BIM objects for construction works	<i>WG 11 had their first meeting November 8th 2024. The current work item involves standardizing the use of BIM object in the built environment. The work is based on the Norwegian Standard NS 8360-series that has been used in Norway since 2015</i>
CEN/TC 442/WG 12	Digitalization of construction products performance characteristics	<i>WG 12 had their first meeting 2024-10-29 in Brussels. Targets how to use CPR and Digital Product Passports DPP System of construction Products.</i>

Technical Committee 442/WG9 WI → Case Studies TR

WI 00442049 CEN/TR
18077:2024 Digital twins
in built environment -
Use cases (Published)

<https://www.une.org/encuentra-tu-norma/busca-tu-norma/norma?c=N0073780>



UNE-CEN/TR 18077:2024 (Ratificada)

 Modelado de la información de construcción. Gemelos digitales aplicados al entorno construido. Casos de uso (Ratificada por la Asociación Española de Normalización en diciembre de 2024.)

 Building information modelling - Digital twins applied to the built environment - Use cases (Endorsed by Asociación Española de Normalización in December of 2024.)

 Modélisation des informations de la construction - Jumeaux numériques appliqués à l'environnement bâti - Cas d'usage (Entérinée par l'Asociación Española de Normalización en décembre 2024.)

Fecha ratificación: 2024-12-01 / **Vigente**

ICS: 35.240.67 / Aplicaciones de las tecnologías de la información en la industria de la edificación y de la construcción

CTN: CTN 332 - Digitalización de la información para edificación y obra civil

Equivalencias internacionales: CEN/TR 18077:2024 (Idéntico)

WI 00442049 CEN/TR 18077:2024 Digital twins in built environment - Use cases (Published)

#	Name	General info	Main use	Asset type	Phase
1	D2EPC THESS	Building Residential	Energy Performance	Building Residential	Operation
2	D2EPC NICOSIA	Building School	Energy Performance	Building Tertiary	Operation
3	PLANON DT	Smart Climatized Asset/Space Management	Energy Performance, Control, Events, Space Management	Building Tertiary	Operation
4	SAMBA	Building Office, Coworking	Operation	Building Tertiary	Operation
5	H2 ELECTRO	DT For Technical Marketing	Marketing	Others (Machinery)	Production (Offsite Construction)
6	BRIDGE WEBGL	Bridge INFRA	Maintenance, MARKETING	Civil Infrastructure (Linear)	Operation
8	CRANE	Building Crane	Planning Construction Operations	Others (Machinery)	Operation
9	KUBIK	Building	Test Lab	Building Industrial	Design
10	BRIDGE ZUBIOTE	Bridge INFRA	TEST BRIDGE	Civil Infrastructure (Linear)	Design
13	BIM2TWIN	Digital Twin Of Construction Execution	Construction Management	Building Tertiary	Execution (Onsite Construction)
14	ENERGY_TWIN	Building Office	Commissioning, Operation	Building Tertiary	Commissioning
15	BRIDGE BAsT	Bridge Rail/Road Infra	Operation, Maintenance	Civil Infrastructure (Linear)	Operation
16	ROAD TU	Road Infra	Operation	Civil Infrastructure (Linear)	Operation
20	SNCF	Rail Infra	KNOWLEDGE OPTIMISATION	Civil Infrastructure (Linear)	Operation
21	ZADAR	Airport Infra	Maintenance	Civil Infrastructure (Punctual)	Operation
22	AVILES PORT	Port Infra	Operation, logistics	Civil Infrastructure (Punctual)	Operation

WI 00442049 CEN/TR 18077:2024 Digital twins in built environment - Use cases (Published)

#	Name	General info	Main use	Asset type	Phase
23	SETEC-StMALO	Port Infra	Heritage	Civil Infrastructure (Punctual)	Reconstruction/preservation
25	ETSICCP	Building School	Operation	Building Tertiary	Operation
26	IRRIGATION	Smart Irrigation Management	Operation	Utility network	Operation
27	LLOBREGAT	BUILDING (Council)	Operation	Building Tertiary	Operation
28	HIDROPOWER	Powerplant Rabenstein (AUSTRIA)	Maintenance, optimization, simulation and VR	Energy Infrastructure	Operation
29	NUCLEAR_DECOMM	(Confidential)	Optimization, Simulation, Training	Building Industrial	Dismantling
30	ST_ETIENNE (see note 1)	Building School	Optimization, simulation, operation	Building Tertiary	Operation
31	ECOLE CENTRALE	Building School	BIM model, CMMS and BMS DATA.	Building Tertiary	Operation
32	LILLE	Building Public	Facilities ticketing system interface	Building Tertiary	Operation
33	TNO SPHERE	Building Residential	Energy management	Building Residential	Operation
34	VTT SPHERE	Abloy factory	Energy optimization	Building Tertiary	Operation
35	EGIS	Gironde Estuary, Grand Port Maritime de Bordeaux	Operation and maintenance	Civil Infrastructure (Punctual)	Operation
Note 1: Approval pending by AFNOR					



2023: 1st NWI

Example of Case Study

DTWIN Puerto de Avilés

CEN/TC 442/WG 9

1 Use case: DIGITAL TWIN FOR SEA PORTS (SMART AND CONNECTED)

1.1 General information

Typology: Critical transport infrastructure

Location: Avilés and Barcelona Port Authorities

Asset owner: Avilés Port Authority

Building Digital Twin (BDT) manager: Sergio Velázquez Cuervo

1.2 Main use of the DT

A Port DT is a platform that integrates a set of web services that help to visualize and manage assets, infrastructure and superstructure through the link between the geometric and geographic representation of BIM/GIS and the industrial data, investment plans, new assets, files of external and new projects within the port infrastructure, together with the administrative information managed by the existing solutions in terms of maritime traffic management, port community systems, occupancy level and planning of operations, maintenance and operation and compliance with environmental regulations.



Fig. 1. Digital Twin Platform for the port terminal allocation management, including customer and other public and private data for the *asset*.

Integration of information about port assets, managed by the Dynamics NAV management software (through SDAP web services, published by the same software) with the geometric identification of each asset represented in the GIS viewer, query filters are added and search that allow the user to interact with the information and geometry, contains other functionalities that allow viewing the detailed information of the asset, downloading the filters made into Excel files and navigating to the management software through links managed in the DT platform and NAV DYNAMICS.

1.3 Description of the DT

Currently, seaports handle more than a half of all goods traded between the EU-28 and the rest of the world. To this respect, a port DT will help maritime and logistics industry to be:

CEN/TC 442/WG 9

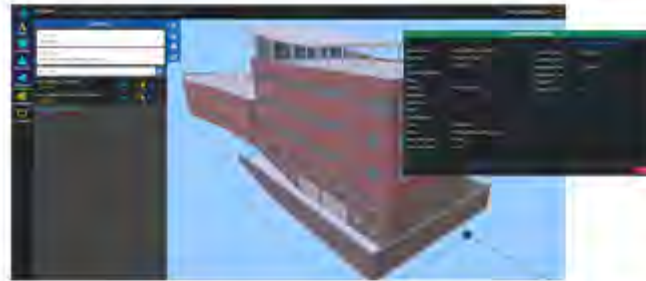


Fig. 2. Integration of DIGITAL TWIN as a service for the management and maintenance of building assets, industrial plants, treatment plants, etc. Administrative building of Avilés Port Authority.

The asset inventory modules, preventive and corrective maintenance management of the asset and the link with the asset monitoring system controlled through IoT technology managed by AGATA are incorporated as part of the solution.

1.5 Replication potential

A port DT is a vehicle to attain the maximum post-project uptake of the Asset Digital Twin foreground, and its impact on the widest possible audience of EU Ports Authorities, Terminal Operators and Multimodal Logistics Experts (Researchers), as well as more general audiences either in the Logistic Sector or Public Administration. Although each port infrastructure is different, the replicability of the Port DT is feasible because it can be implemented as a trustworthy and accurate digital representation of the critical infrastructure and its systems. It can create feedback loops that can enable near real-time data transfer between the virtual replicate and critical infrastructure systems including simulated data series, the creation of a common data environment, implement data analytics, including machine learning and simulation which can provide predictive tools "what-if" situations.

1.6 Relevant links

1. The Digital Twin of Port of Rotterdam: <https://www.dinque.com/media/111499999>

2. Analytics with digital twinning: A decision support system for maintaining a resilient port. Chenhao Zhao, Jie Xu et al. Decision Support Systems 143 (2021) 113496.

3. Multi-sector applications and development challenges of digital twin-driven management in global smart ports. Kai Wang et al. Case Studies on Transport Policy, Volume 9, Issue 3, September 2021, Pages: 1298-1312.

1.7 Contact information

Name, email and telephone of a contact person on behalf of the DT owner and/or the BDT manager. This information will not be included in the final TR, but will be used to ask for further information, if needed.

WI 00442049 CEN/TR 18077:2024 Digital twins in built environment - Use cases (Published)

	USE CASE	MAIN USE #	SECONDARY USE #
1	Optimization Design		16
2	Optimization Construction	8,13	
3	Optimization Commissioning	14	
4	Optimization Operation (Energy Performance)	1,2,14,30,33,34	3,4,27,31
5	Optimization Operation (Space and Administrative Management)	3,22,25,27,32	4,20,30,31
6	Optimization Operation (Logistics)	35	20,22
7	Optimization Operation (Heritage)	23	
8	Optimization Maintenance	15,16,20,21,28,31	6,22,25,27
9	Optimization Dismantling	29	16
10	Safety	4	29
11	Training		4,28,29
12	Marketing	5,6	
13	Test Lab	9,10	
14	Others	26	34,35

All the case studies have been grouped following similar characteristics to extract representative guidelines of the examples received, with dedicated focus on extracting potential use cases.

WI 00442055 prEN 18162 Building information modelling - Digital twins applied to the built environment — Concept and definitions (*Enquiry open 2025-02-06*)

Documents to consider



CEN TC442/WG9: 8th Meeting (On-line)
New Working Item proposal

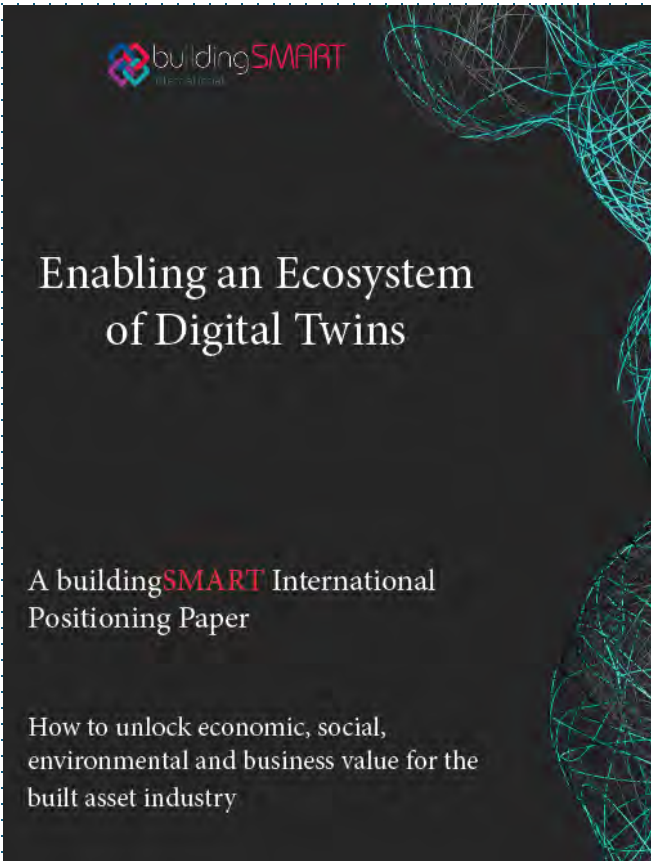
20th October 2023

WI 00442055 prEN 18162 Building information modelling - Digital twins applied to the built environment — Concept and definitions (*Enquiry open 2025-02-06*)

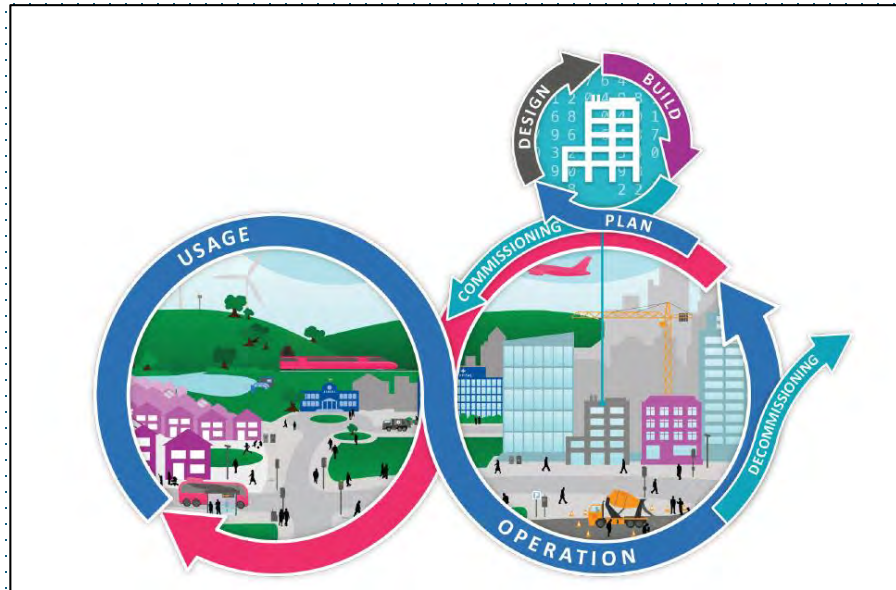
Documents to consider

- SPHERE/BDTA White Paper 1 (2019)
- BuildingSmart Position Paper (2020)
- Digital Twin Construction paper (2020)
- ISO 23247-1-2021 (2021)
- ISO 30373

Documents to consider



- SPHERE/BDTA White Paper 1 (2019)
- **BuildingSmart Position Paper (2020 & 2023 update)**
- Digital Twin Construction paper (2020)
- ISO 23247-1-2021 (2021)
- ISO 30373



Documents to consider

- SPHERE/BDTA White Paper 1 (2019)
- BuildingSmart Position Paper (2020 & 2023 update)
- **Digital Twin Construction paper (2020)**
- ISO 23247-1-2021 (2021)
- ISO 30373

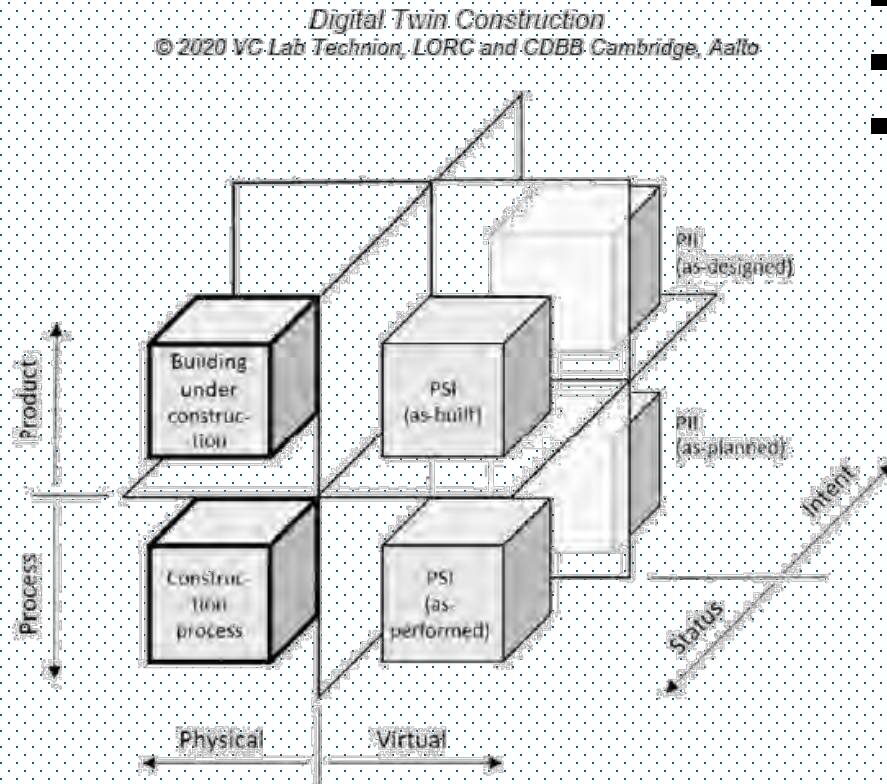


Figure 5. Information aspects of the domain within the three-dimensional space defined by the digital – physical, the product – process and the ex-ante and ex-post dimensions.

Digital Twin Construction
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Digital Twin Construction

Rafael Sacks, Technion IIT

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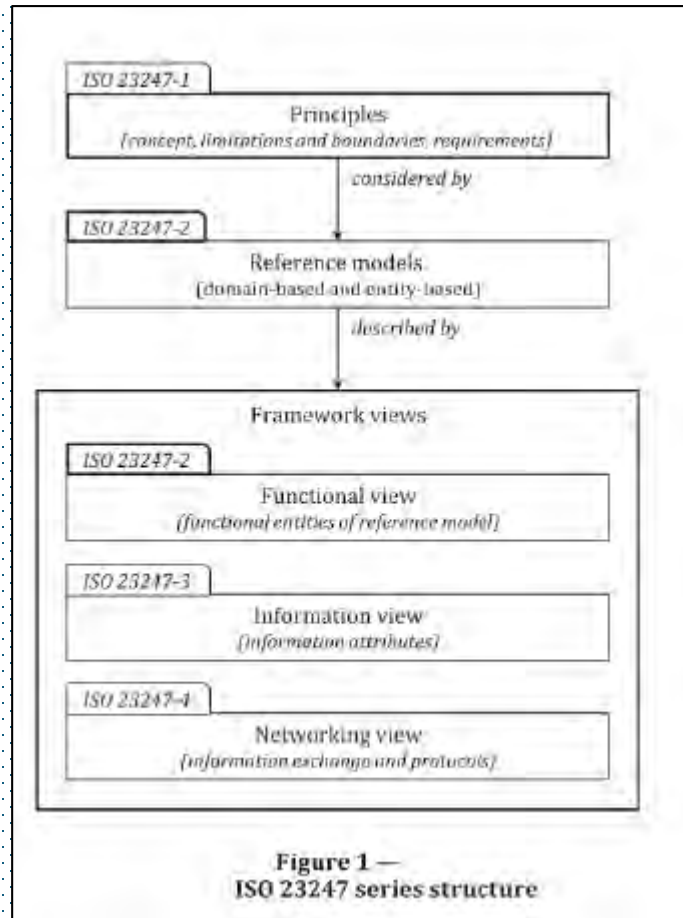
Ergo Pikas, Aalto University

Sally Xie, University of Cambridge

Mark Girolami, University of Cambridge

Documents to consider

- SPHERE/BDTA White Paper 1 (2019)
- BuildingSmart Position Paper (2020 & 2023 update)
- Digital Twin Construction paper (2020)
- **ISO 23247-1-2021 (2021)**
- ISO 30373



Automation systems and integration — Digital twin framework for manufacturing

Part 1: Overview and general principles

Documents to consider



- SPHERE/BDTA White Paper 1 (2019)
- BuildingSmart Position Paper (2020 & 2023 update)
- Digital Twin Construction paper (2020)
- ISO 23247-1-2021 (2021)
- **ISO 30373 (2023)**

3.1.1 digital twin DTw

digital representation (3.1.8) of a *target entity* (3.1.3) with data connections that enable convergence between the physical and digital states at an appropriate rate of synchronization

Note 1 to entry: Digital twin has some or all of the capabilities of connection, integration, analysis, simulation, visualization, optimization, collaboration, etc.

Note 2 to entry: Digital twin can provide an integrated view throughout the life cycle of the target entity.

WI 00442055 prEN 18162 Building
information modelling - Digital twins
applied to the built environment —
Concept and definitions (*Enquiry open*
2025-02-06)

CEN/TC 442/WG 9

Date: 2024-11

prEN XXXXX:2024

Secretariat: UNE

**Building information modelling — Digital twins applied to the built
environment — Concept and definitions**

CCMC will prepare and attach the official title page.

WI 00442055 prEN 18162 Building information modelling - Digital twins applied to the built environment — Concept and definitions (*Enquiry open 2025-02-06*)

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WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442/WG9 Liaisons

ISO/IEC JTC 1/SC 41

Manufacturing sector has strongly embraced the concept in their industry 4.0 approach, reaching the first ISO in the matter

manufacturing sector has strongly embraced the concept in their industry 4.0 approach, reaching the first ISO in the matter → ISO 23247

Abstract

[Preview](#)

This document provides an overview and general principles of a digital twin framework for manufacturing including:

- terms and definitions;
- requirements of the digital twin framework for manufacturing.

General information

Status :  Published

Publication date : 2021-10

Edition : 1

Number of pages : 12

Technical Committee : ISO/TC 184/SC 4 Industrial data

ICS : 25.040.40 Industrial process measurement and control | 35.240.50 IT applications in industry

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442/WG9 Liaisons

Currently working on
a Reference
Architecture →
ISO/IEC 30188 Digital twin
RA

NEW
PROJECT

ISO/IEC AWI 30188

Digital Twin — Reference architecture

Under development

Drafting has started.

WORKING GROUP #1: BDTE DEFINITIONS & ROLES

Technical Committee 442/WG9 Liaisons

Currently working on
a Reference
Architecture →
SC41/AG31 report
Guidance on the
integration of IoT and
digital twin in data spaces

This document provides guidance on the integration of IoT systems and digital twins in data spaces.
PWI JTC1-SC41-17

GENERAL INFORMATION

Status: 🔄 Under development

Publication date :

Working Groups :

Category: [Citiverse](#)

SDO:
[ISO](#)

Latest published version :

https://www.iec.ch/dyn/www/f?p=103:38:9406018951756::::FSP_ORG_ID,FSP_APEX_PAGE,FSP_PROJECT_ID:20486,23,118815

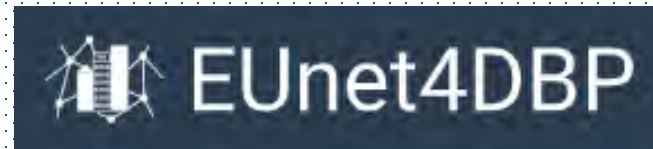
WORKING GROUP #2: OPEN TECHNOLOGY FRAMEWORK

Ambition: Generate an Open Technology Framework which will allow an improved interoperability across Building Digital Twins.

The technology backed by BDTA will follow our guiding principles to advance towards the proposed Vision and Mission.

Whenever possible, developments will consider external Open Standards.

Open knowledge generated in parallel EU projects welcomed!!!



WORKING GROUP #2: OPEN TECHNOLOGY FRAMEWORK

10/2021 → Kick off meeting as
Coorganised Event with ECTP



<https://www.cibw78-ldac-2021.lu/programme/side-events/>

Thursday, 14 October 2021 - Workshop Linking EU H2020 projects on digitization in the construction and maintenance industry: Linked Data and ontologies for BIM and Building Digital Twins

Date: Thursday, 14 October 2021

Room: José Ensch

Chaired by Mathias Bonduel, Neanex Technologies, and Pierre Bourreau, Nobatek/INEF4

In at least five finalizing European H2020 research projects** focusing on digitization in the building industry, Linked Data technologies are actively applied to tackle well-known interoperability challenges in the sector. Naturally, this resulted in the creation of a series of technical deliverables (ontologies, data modelling patterns constraints, processes, tools, etc) with both similarities and differences between the different ongoing projects. During the workshop, each of the five projects will share their technical insights on the subject through short technical presentations. The audience might expect to get a view on the reasons for applying Linked Data, the progress made during the project and the lessons learned. At the end of the presentations a roundtable discussion will be held between the different projects to discuss how the Linked Data related outcomes of the different projects can support each other and persist through the BDTA after the independent projects have ended.

** H2020 research projects involved in the workshop:

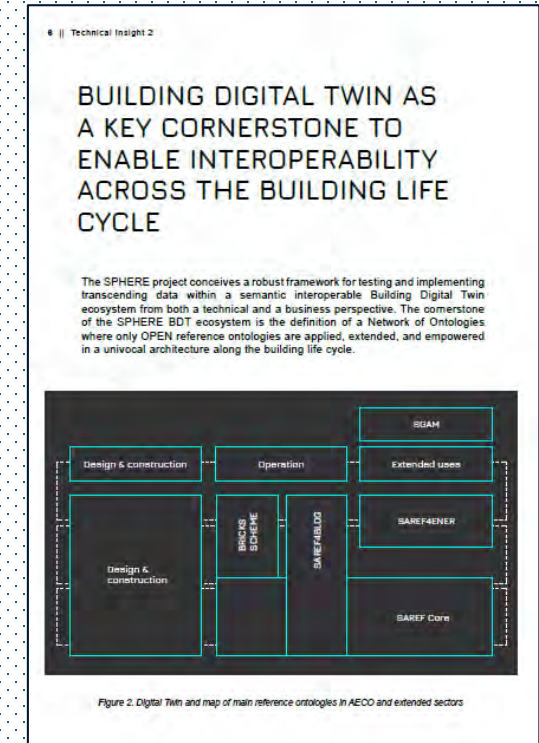
- SPHERE: <https://sphere-project.eu/>
- BIM4EEB: <https://www.bim4eeb-project.eu/>
- BIM-SPEED: <https://www.bim-speed.eu/en>
- BIM4REN: <https://bim4ren.eu/>
- BIMERR: <https://bimerr.eu/>



WORKING GROUP #2: OPEN TECHNOLOGY FRAMEWORK

PUBLICATIONS 2022

Technical Insight #2 (published in May 2022)



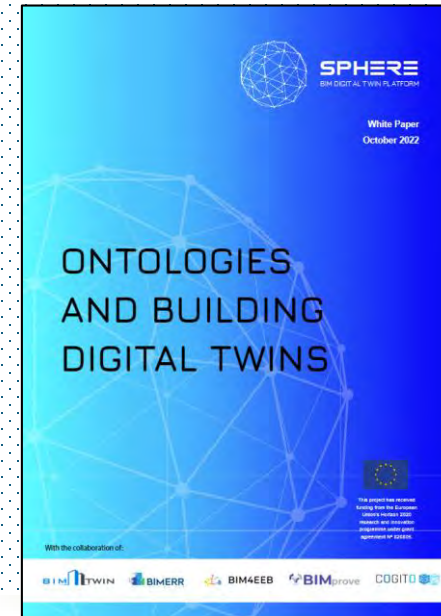
WORKING GROUP #2: OPEN TECHNOLOGY FRAMEWORK

PUBLICATIONS 2022

White Paper #3

Ontology Network for BTDEs

In collaboration with 5 sister projects!!!



BIM4EEB



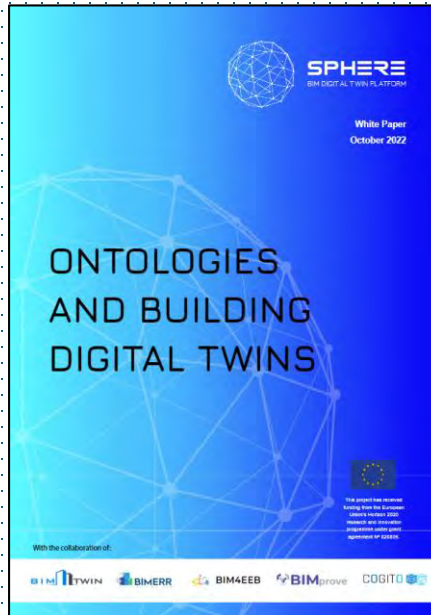
BIMprove



WORKING GROUP #2: OPEN TECHNOLOGY FRAMEWORK

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Building digital twin as a key cornerstone to enable interoperability across the building life cycle		BIMprove, safety-critical situation detection	
Future direction: towards a unified reference architecture for building life cycle management		IDP digital twin platform	
Alignment and interoperability		BIM4EEB	
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The World Wide Web: A very short personal history		NEANEX material passports and portal	
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3.1 BUILDING			

WORKING GROUP #2: OPEN TECHNOLOGY FRAMEWORK



The class `bot:Interface` is used to describe the relationship between some specific zones and elements in detail. The concept of `bot:Interface` is useful in different situations:

- the heat transmission area of the surface between a space and an adjacent wall can be used to determine the heat loss from that space through this wall;
- the localisation of the intersection between a pipe and a wall can be used to specify where to apply fire sealing;
- the type of access between two zones can be used to specify access restrictions for use in door navigation.

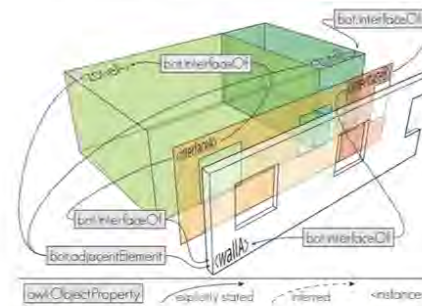


Figure 16: Interfaces between two zones and a wall. Inter faces can be used to qualify (i.e., attach additional information to) topological relationships between zones, elements, or zones and elements.

Geometry and properties are outside the scope of BOT but there are ways to integrate them as well.

BOT is designed to function as a central element in the interdisciplinary communication. In addition, it aims at being the key entry point to connect the construction sector to adjacent domains.

BIM4EEB Digital Construction Ontologies, DiCon⁴⁷

Digital Construction Ontologies (DiCon) were developed in the BIM4EEB based on the earlier ontology work carried out in the DiCtion project in Finland. DiCon is composed of a set of interrelated ontology modules that aim to capture the different aspects of construction and renovation projects. Since there are significant standardized data models and other ontology work in this domain – as reported in this white paper – the objectives of DiCon have been to link to and integrate with these other models.

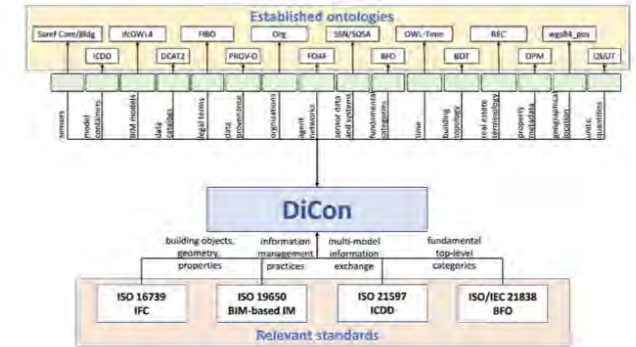


Figure 17: Relationships of DiCon with relevant standards and established ontologies

Figure 17 shows the relationship of DiCon ontologies with the relevant standards and established ontologies related to construction domain. As technical solutions for integration the DiCon uses the following:

- A top-level ontology for the uniform categorization of concepts in the different modules of DiCon.
- Alignment modules that establish the correspondences of DiCon terms with the terms of other ontologies. All the references to external concepts are placed in the alignment modules (except for instance vocabularies).

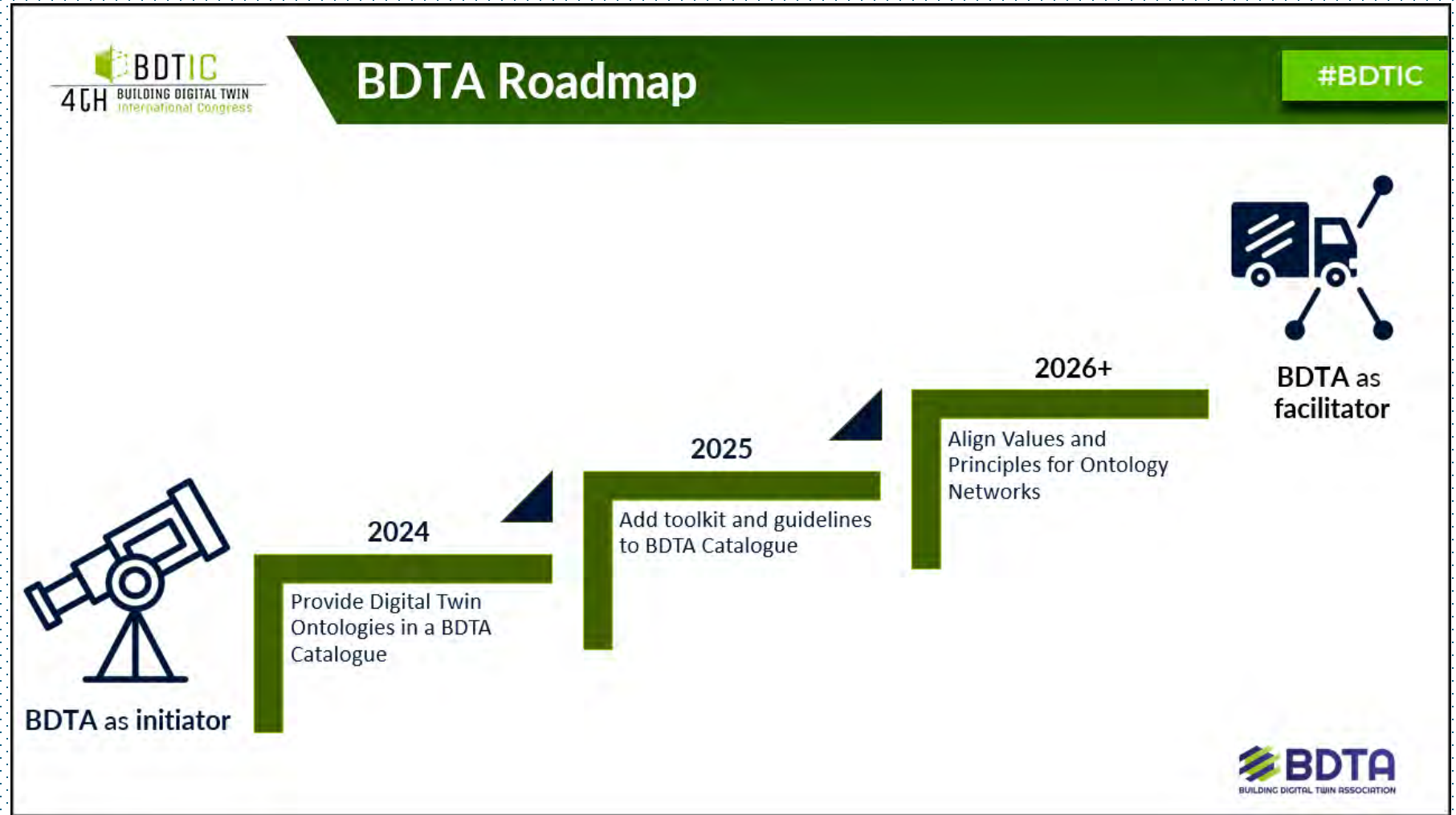
As shown in Figure 17, the standards considered most relevant for DiCon are:

- ISO 16739 Industry Foundation Classes (IFC): The universally supported BIM standard, according to which BIM models are used in BIM4EEB and DiCon.
- ISO 19650 BIM-based information management: A process standard whose relevant terminology is formalized in the DiCon.
- ISO 21597 Information Container for Linked Document Delivery (ICDD): A standard for the exchange of packages of interrelated documents or models.
- ISO/IEC 21838 Basic Formal Ontology (BFO): A standard top-level ontology used in DiCon.

DiCon is organized into several modules, shown in Figure 18 as blue boxes, where the arrows between the blue boxes represent the import relations (`owl:imports`) between the modules.

WORKING GROUP #2: OPEN TECHNOLOGY FRAMEWORK

First Roadmap
agreed in BDTIC4
during the
7th Workshop of the
WG2



WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

Chronology:

04/2024 → WG2 Meeting during BDTIC4: Ontology Network Catalog Proposed by the experts

07/2024 → BDTA Internal Project generated by SEMMTECH and accepted by the board

02/2025 → First publication of the BDTA's OntNetCatalog

Expected 04/2025 → End of Phase I



WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

Driving the future of BDT : The launch of this platform reinforces BDTA's mission to lead open advancements in the *Building Digital Twin* domain.

Ontology Catalogue

The BDTA Ontology Catalogue platform aims to ensure building digital twin ontologies can be shared, explored, accessed and used. This catalogue serves as a central hub for discovering ontologies relevant to the building digital twin domain, helping to improve collaboration, reuse and exchange of knowledge across the domain and industry. See the About page for more information.



WORKING GROUP #2: Ontology Network Catalog





[Home](#)
[Ontologies](#)
[About](#)
[Contact](#)

Ontologies

entries per page

Search:

Name	Website	Description	License	Domain	Release date
AEC3PO: Architecture, Engineering, Construction Compliance Checking and Permitting Ontology	Website	AEC3PO is designed to model compliance and permitting processes in the construction industry, on ...			19/12/2023
BPO: Building Product Ontology	Website	The Building Product Ontology defines concepts to describe (building) products in a schematic way...	CC BY 4.0		13/01/2025
Brick Ontology	Website	Brick is an open-source effort to standardize semantic descriptions of the physical, logical and ...	BSD		13/01/2025
Building Topology Ontology	Website	The Building Topology Ontology (BOT) is a minimal ontology for describing the core topological co...	W3C Community Contributor License Agreement (CLA)		13/01/2025
Digital Construction Ontologies (DiCon)	Website	Digital Construction Ontology - semantic interoperability between systems in the construction and...	CC BY 4.0		01/07/2022

WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

In 2025: End of PHASE I...

Ontologies Catalogue

- A comprehensive directory of Ontologies in the BDT domain
- Facilitate easy search of Ontologies in the BDT domain
- Enable users to understand the scope, purpose and structure of available ontologies



WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

... and start of PHASE II

Toolkit and guidelines

- Ontology adoption and interaction supported for various applications and projects
- Improved collaboration among software providers, ontology developers and users
- Basic training and guidelines available for users



WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

PHASE I – 2025 – Final Steps

Screen and include ontologies related to three key

Use Cases extracted from CEN TR18077:2024 conclusions



UNE-CEN/TR 18077:2024

🇪🇸 Modelado de la información de construcción. Gemelos digitales aplicados al entorno construido. Casos de uso (Ratificada por la Asociación Española de Normalización en diciembre de 2024.)

🇬🇧 Building information modelling - Digital twins applied to the built environment - Use cases (Endorsed by Asociación Española de Normalización in December of 2024.)

🇫🇷 Modélisation des informations de la construction - Jumeaux numériques appliqués à l'environnement bâti - Cas d'usage (Entérinée par l'Asociación Española de Normalización en décembre 2024.)



WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

PHASE I – 2025 – Final Steps

Screen and include ontologies related to
three key Use Cases extracted from

CEN TR18077:2024 conclusions

1- Building Digital Twin Operation & Maintenance (O&M) Optimisation

Use cases IV, V, VI, VII and VIII

I	Design Optimization
II	Construction Optimization
III	Commissioning Optimization
IV	Operation Optimization (Energy Performance)
V	Operation Optimization (Space and Administrative Management)
VI	Operation Optimization (Logistics)
VII	Operation Optimization (Waste Management)
VIII	Maintenance Optimization



WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

PHASE I – 2025 – Final Steps

Screen and include ontologies related to
three key Use Cases extracted from

CEN TR18077:2024 conclusions

2 - Building Digital Twin Prototype Optimisation

Use cases I, II, III, IV, and VI

I	Design Optimization
II	Construction Optimization
III	Commissioning Optimization
IV	Operation Optimization (Energy Performance)
V	Operation Optimization (Space and Administrative Management)
VI	Operation Optimization (Logistics)
VII	Operation Optimization (Waste Management)
VIII	Maintenance Optimization



WORKING GROUP #2: ONTOLOGY NETWORK CATALOGUE

PHASE I – 2025 – Final Steps

Screen and include ontologies related to
three key Use Cases extracted from

CEN TR18077:2024 conclusions

3 - Digital Building Permits

Use cases I, II, III

I	Design Optimization
II	Construction Optimization
III	Commissioning Optimization
IV	Operation Optimization (Energy Performance)
V	Operation Optimization (Space and Administrative Management)
VI	Operation Optimization (Logistics)
VII	Operation Optimization (Waste Management)
VIII	Maintenance Optimization



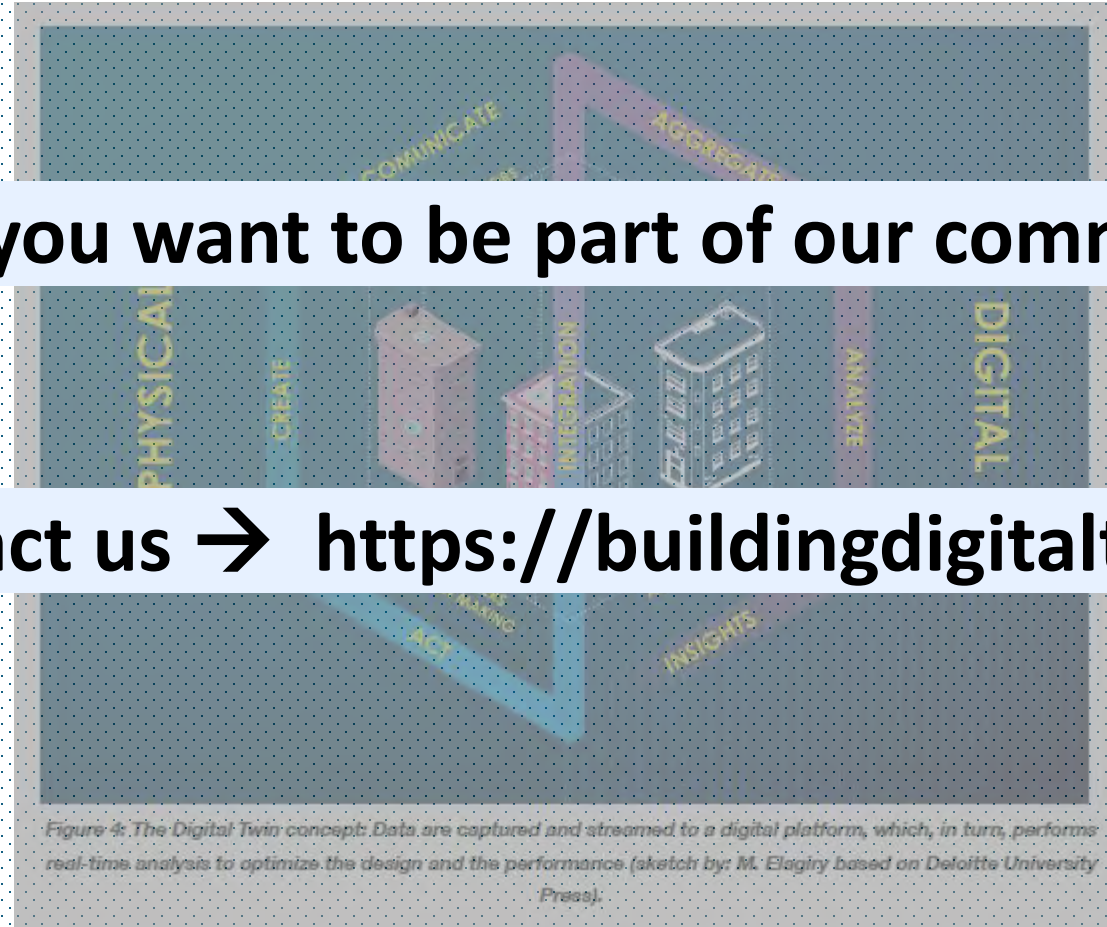
BARCELONA SCEWC 2025

BDTA – DTC Workshop
Critical Civil Infrastructures
Railway, Roads
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Water & Energy utilities



Do you want to be part of our community?

Contact us → <https://buildingdigitaltwin.org/>



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Questions?

